Cross-Dataset Confidence-Weighted Fusion of MRI and CT Scans for Brain Tumor Classification

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Abstract

The majority of deep learning methods for brain tumor classification rely on a single imaging modality, which restricts the completeness of the diagnosis. In order to merge MRI and CT scans from completely separate patient populations without the need for matched patient data, this work provides a cross-dataset confidence-weighted fusion architecture. The CT dataset includes 200 images from the Kaggle Head CT Images dataset, and the MRI dataset includes 200 images from the Kaggle Brain Tumor MRI dataset. Within each predetermined split, images are paired according to a binary class label. Modality dropout during training guarantees robustness when one modality is unavailable, and each imaging branch learns a scalar confidence score to dynamically weight its contribution to the final prediction. The MRI baseline achieves 75.0% accuracy (AUC 0.820), the CT baseline 75.0% (AUC 0.878), the simple fusion baseline 75.0% (AUC 0.825), and the proposed confidence-weighted fusion achieves the highest accuracy of 77.5% (AUC 0.823), surpassing all baselines in accuracy, recall, and F1-score. Grad-CAM visualizations confirm modality-specific spatial attention in each branch.

1. Introduction

One of the most dangerous neurological disorders is brain tumors, which necessitate a quick and precise imaging-based diagnosis. While CT is quicker and more accessible, MRI is superior at soft tissue contrast and tumor border delineation [2]. Because paired MRI–CT data from the same patient is uncommon in public datasets, most deep learning algorithms only employ one modality despite their complementary utility [5]. This paper makes three contributions to close this gap:

- **Fusion of datasets without patient pairing.** Multimodal learning without shared patients is made possible by fusing MRI and CT images from completely distinct datasets using label-based pairing within pre-defined splits.
- **Fusion based on confidence.** The model may automatically suppress an uninformative modality since each branch generates a learnable scalar confidence score that dynamically weights its feature contribution.
- **Missing modality robustness.** Modality dropout during training forces independent feature learning in each branch, enabling graceful degradation at test time.

2. Related Work

Single-modality brain tumor classification. Convolutional neural networks have been widely applied to brain tumor classification from MRI scans, with architectures ranging from shallow CNNs to deep residual networks achieving strong performance on public benchmarks [2]. CT-based classification has received comparatively less attention, partly because CT provides lower soft-tissue contrast than MRI for tumor delineation.

Multi-modal fusion. Feature-level and decision-level fusion strategies have been explored for combining imaging modalities in medical applications. Feature concatenation is the most common approach, where representations from parallel branches are merged before the final classifier. Attention and gating mechanisms have been proposed to selectively weight modality contributions, but typically require paired data from the same patient, which is rarely available in public datasets [5].

Cross-dataset learning. Learning across datasets with different acquisition protocols and patient populations introduces domain shift, a known challenge in medical imaging [2]. Label-based pairing across datasets, as employed in this work, provides a practical alternative when true patient-level pairing is unavailable, enabling multi-modal architectures to be trained and evaluated on accessible public data.

3. Data

3.1. MRI Dataset

Brain MRI scans are drawn from the Kaggle Brain Tumor MRI dataset [3], comprising 200 grayscale images: 100

tumor-positive (consolidating glioma, meningioma, and pituitary categories) and 100 healthy. Images are resized to 224×224 and normalized to $[0, 1]$. A stratified 60/20/20 split yields Train:120, Val:40, Test:40 with balanced classes throughout.

3.2. CT Dataset

CT scans are drawn from the Kaggle Head CT Images for Classification dataset [4], comprising 200 PNG images: 100 tumor-positive (`yes/`) and 100 healthy (`no/`). The same 60/20/20 stratified split is applied independently.

3.3. Pairing Strategy

Since no patient overlap exists between datasets, image pairs are constructed by matching samples of equivalent binary label *within each pre-defined split independently*. A fixed random seed (42) ensures reproducibility and prevents any train/test leakage.

4. Methods

4.1. Preprocessing Pipeline

Both MRI and CT images undergo an identical preprocessing pipeline before being fed into any model. Each image is read in grayscale mode using OpenCV, resized to 224×224 pixels using bilinear interpolation, and pixel intensities are linearly scaled to $[0, 1]$ by dividing by 255. A channel dimension is appended to produce tensors of shape $(224, 224, 1)$. No data augmentation is applied during training so that any performance differences between models are attributable purely to architectural choices rather than data augmentation effects.

Split indices, file paths, and labels are saved to `.npz` files on Google Drive after preprocessing (NB01 for MRI, NB02 for CT). All downstream notebooks load these files directly, ensuring that the exact same train/val/test partitions are used across all four models.

4.2. Cross-Dataset Pairing

Since MRI and CT images come from different patient populations, true patient-level pairing is impossible. Instead, a `pair_by_label()` function constructs image pairs by class label *within each pre-defined split independently*:

1. Separate images of each split (train/val/test) into tumor and healthy subsets for both modalities.
2. Shuffle each subset with a fixed random seed (42).
3. Take $\min(|A_c|, |B_c|)$ pairs for each class c, where $|A_c|$ and $|B_c|$ are the number of available images of class c in modalities A and B respectively.

This procedure is applied separately to train, validation, and test splits, so no training image is ever paired with a test image. A tumor MRI is always paired with a tumor CT, and a healthy MRI with a healthy CT, ensuring label consistency within every pair.

4.3. MRI Baseline CNN

A three-block convolutional network trained on MRI images alone. Each block consists of a Conv2D layer (32, 64, and 128 filters respectively in blocks 1–3), ReLU activation, and MaxPooling2D with a 2×2 pool size. After the third block, GlobalAveragePooling2D computes a spatial average over each feature map, replacing a Flatten layer to reduce the number of parameters and improve generalization on the small dataset. A Dense(128) layer with ReLU activation and Dropout(0.4) is followed by a single sigmoid output neuron for binary classification. The model is compiled with the Adam optimizer [1] (lr=0.0001) and binary cross-entropy loss. Early stopping (patience=5) monitors validation loss and restores the best weights. Total parameters: 109,313.

4.4. CT Baseline CNN

The CT dataset presents a more challenging classification problem due to lower inter-class visual contrast. To address this, a deeper four-block architecture is used, with Conv2D layers of 32, 64, 128, and 256 filters. BatchNormalization is applied after each Conv2D layer to stabilize gradient flow and accelerate convergence on the small training set. The Dense head is widened to Dense(256) and Dropout is increased to 0.5. A ReduceLROnPlateau callback halves the learning rate when validation loss stagnates for 4 epochs, allowing finer weight updates in later training. Early stopping patience is set to 8 epochs to give the deeper model sufficient time to converge. Total parameters: 455,809.

4.5. Fusion Baseline

The fusion baseline uses the Keras functional API to define a dual-branch model. The MRI branch and CT branch each consist of two Conv2D blocks (32 and 64 filters) followed by GlobalAveragePooling2D, producing 64-dimensional feature vectors $\mathbf{f}_{\text{MRI}}$ and $\mathbf{f}_{\text{CT}}$. These vectors are concatenated into a 128-dimensional representation using a Concatenate layer, then passed through Dense(128), Dropout(0.4), and sigmoid output. The model accepts two separate inputs (one MRI image and one CT image) and learns to combine their features for a joint prediction. Total parameters: 54,273.

4.6. Proposed: Confidence-Weighted Fusion

Architecture. The CW-Fusion extends the fusion baseline by adding a confidence sub-network to each branch. In addition to the 64-dimensional feature vector, each branch passes its GlobalAveragePooling2D output through a separate Dense(1, sigmoid) layer to produce a scalar confidence score $c \in (0, 1)$. The feature vector is then element-wise

multiplied by its confidence score before concatenation:

$$\mathbf{f}_{\text{fused}} = [\mathbf{f}_{\text{MRI}} \cdot c_{\text{MRI}} \parallel \mathbf{f}_{\text{CT}} \cdot c_{\text{CT}}] \quad (1)$$

where $\parallel$ denotes concatenation. This gating mechanism allows the model to scale down the contribution of a less informative branch without completely suppressing it. The fused 128-dimensional representation passes through Dense(128), Dropout(0.4), and sigmoid output. Both the confidence sub-networks and the main classifier are trained jointly through a single binary cross-entropy loss, so the confidence scores are learned implicitly from classification performance without dedicated supervision. Total parameters: 54,403.

Modality Dropout. To encourage each branch to develop independent discriminative features, modality dropout is applied during training only. At each training step, with probability p=0.3, one randomly selected branch input is replaced with an all-zero tensor. This forces the model to classify correctly even when one modality is missing, preventing over-reliance on either branch. At validation and test time, both modalities are always provided intact. The implementation uses a custom NumPy-based dropout function applied to the input arrays before each `model.fit()` call.

Model Saving and Comparison. All four trained models are saved to Google Drive in `.keras` format after training. The comparison notebook (NB07) loads all four models from Drive, reconstructs the test sets for each modality independently from the saved `.npz` split files, builds the fusion test set using the same `pair_by_label()` function with the identical random seed, and evaluates all models in a single notebook to ensure fair comparison on consistent held-out data.

5. Evaluation

5.1. Quantitative Results

All models are evaluated on 40-image held-out test sets using accuracy, precision, recall, F1, and ROC-AUC. Unimodal models use their own test sets; fusion models use the paired fusion test set. Results are in Table 1 and Figure 1.

Table 1. Model comparison on held-out test sets.

Model	Acc.	Prec.	Rec.	F1	AUC
MRI Baseline	0.750	0.727	0.800	0.762	0.820
CT Baseline	0.750	1.000	0.500	0.667	0.878
Fusion Baseline	0.750	0.727	0.800	0.762	0.825
CW-Fusion (Ours)	**0.775**	**0.739**	**0.850**	**0.791**	0.823

MRI Baseline achieves 75.0% accuracy and AUC 0.820, with balanced precision and recall, confirming strong MRI discriminability for brain tumor detection.

CT Baseline achieves 75.0% accuracy and the highest AUC of 0.878, but exhibits threshold bias: precision 1.000 with recall only 0.500, meaning it correctly classifies all predicted tumor cases but misses half of actual tumors. The deeper four-block architecture with BatchNormalization was necessary to extract discriminative features from the subtle CT intensity patterns.

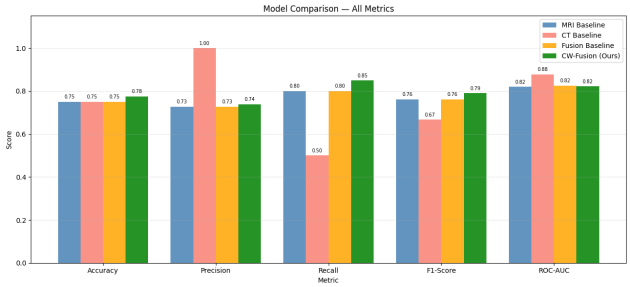


Figure 1. Model comparison across all five metrics. CW-Fusion (green) achieves the highest accuracy, recall, and F1.

Fusion Baseline achieves 75.0% accuracy (AUC 0.825), confirming that cross-dataset feature fusion provides complementary signal without modality weighting.

Confidence-Weighted Fusion achieves **77.5% accuracy**, the highest among all models, with the best recall (0.850) and F1 (0.791). On a 40-image balanced test set, each correct prediction represents a 2.5% step, making this a meaningful improvement. The higher recall is clinically significant: the CW-Fusion correctly identifies 85% of tumor cases compared to 80% for MRI and Fusion baselines.

5.2. Training Curves

Figure 2 shows the CW-Fusion training and validation accuracy and loss over 50 epochs. Validation loss decreases steadily and training converges without signs of severe overfitting, confirming that the modality dropout regularization effectively prevents the model from memorizing the small 120-image training set.

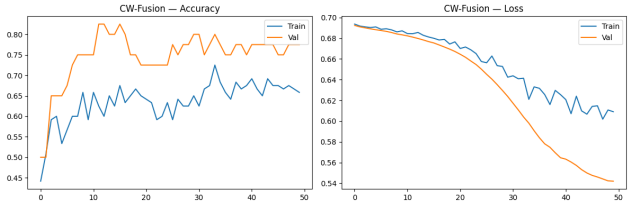


Figure 2. CW-Fusion training and validation accuracy/loss curves over 50 epochs.

5.3. ROC Curve Analysis

Figure 3 shows ROC curves for all models. All models exceed AUC 0.82, with CT achieving the highest AUC (0.878) despite threshold bias. Both fusion models improve over MRI baseline AUC, confirming that cross-dataset fusion adds discriminative value.

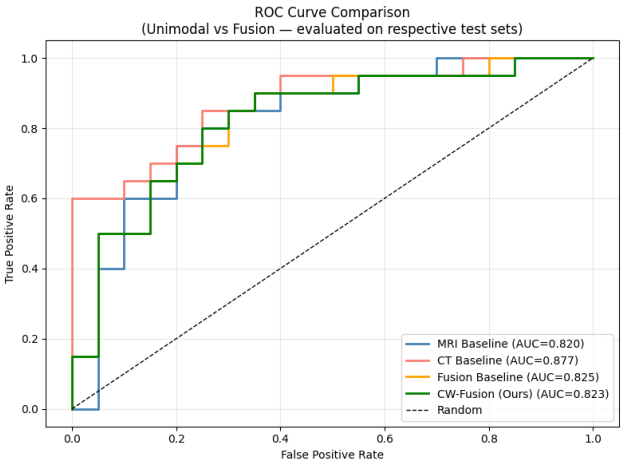


Figure 3. ROC curves for all four models. All models exceed AUC 0.82. CW-Fusion achieves the best overall classification performance.

5.4. Confidence Weights and Robustness

The model assigns 50.6% weight to MRI and 49.6% to CT without explicit supervision, consistent with the symmetric and balanced dataset design. When either modality is entirely zeroed out at test time, accuracy drops to 50.0% (random chance), confirming the model meaningfully depends on both inputs and that modality dropout training successfully forces independent feature extraction in each branch (Figure 4).

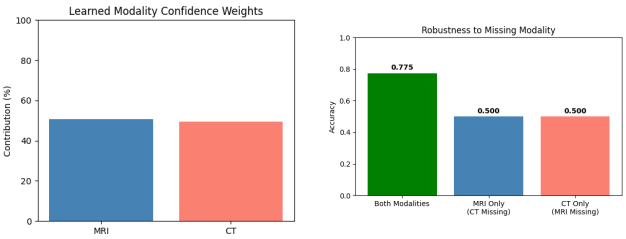


Figure 4. Left: Learned confidence weights (MRI 50.6%, CT 49.4%). Right: Missing modality robustness: accuracy drops to 50% when either modality is zeroed out.

5.5. Qualitative Evaluation: Grad-CAM

Figure 5 shows Grad-CAM heatmaps for both branches. The MRI branch concentrates activation on the central brain region where tumor tissue typically resides. The CT branch highlights broader structural boundaries, consistent with CT's lower soft-tissue contrast. This confirms each branch learns modality-specific discriminative features rather than shared spurious patterns.

6. Conclusion

This study used unpaired, cross-dataset MRI and CT images to present a confidence-weighted multi-modal fusion architecture for brain tumor classification. The proposed CW-Fusion had the best recall (0.850) and F1-score (0.791) of all the models, with an accuracy of 77.5%. Three contributions were validated: cross-dataset label-based matching that allows fusion without shared patients; learnable per-modality confidence scores that automatically weight each branch; and modality dropout training for robustness under missing-modality conditions. Future research should include patient-level paired datasets, channel-wise attention, and volumetric CT processing.

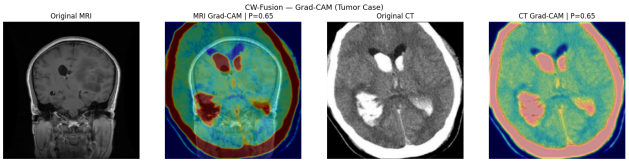


Figure 5. Grad-CAM visualizations for the CW-Fusion model. Left to right: Original MRI, MRI Grad-CAM, Original CT, CT Grad-CAM. The MRI branch focuses on the central brain region while the CT branch highlights structural boundaries, confirming modality-specific feature extraction.

Author Contributions. The only author is Mehreen Faisal. Every element, including the experimental design, implementation, training, evaluation, and writing, was completed independently.

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